

Assignment – Week 6

Introduction to Machine Learning-IIT KGP

Topic: Neural Networks

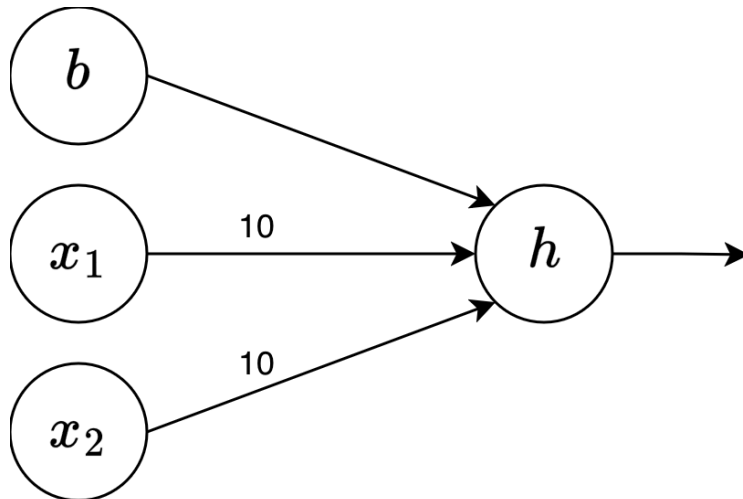
TYPE OF QUESTION: MCQ/MSQ

Number of Questions: 10

Total Marks: $10 \times 2 = 20$

Question 1

Consider a perceptron with two binary inputs x_1 and x_2 . The weight associated with x_1 is $w_1=10$ and the weight associated with x_2 is $w_2=10$ and the bias is $b=-5$. The activation function is a binary threshold function (i.e. $h(z) = 1$, if $z>0$; $h(z) = 0$, otherwise).



Which logical function does the perceptron compute?

- A. AND
- B. OR
- C. XOR
- D. NAND

Correct Answer: B

Detailed Solution: $h = 1$, if $(10x_1 + 10x_2 - 5) > 0$, $h = 0$, otherwise

x1	x2	h (output)
0	0	0
0	1	1
1	0	1
1	1	1

The truth table above shows that the output of the perceptron computes the OR gate.

Question 2

Which of the following operations is performed during backpropagation?

- A. Updating weights using gradients
- B. Computing the forward pass only
- C. Removing hidden layers
- D. Increasing the number of neurons

Correct Answer: A

Detailed Solution: Backpropagation computes gradients and updates weights.

Question 3

Suppose $q=x*y$ and $f=q+z$ where $x=2$, $y=3$ and $z=5$. What is the gradient of f with respect to x , y and z , i.e. $(df/dx, df/dy, df/dz)$?

- A. (3,2,1)
- B. (2,3,1)
- C. (6,6,1)
- D. (3,3,1)

Correct Answer: A

Detailed Solution: $f=xy+z$, so gradients are (3,2,1).

$$df/dx = y = 3, df/dy = x = 2, df/dz = 1$$

Question 4

Which change is most likely to increase the capacity of a neural network?

- A. Increasing hidden neurons
- B. Reducing training samples
- C. Increasing dropout significantly
- D. Reducing layers

Correct Answer: A

Detailed Solution: More hidden neurons increase capacity.

Question 5

Which statement about ReLU is true?

- A. Outputs only between 0 and 1
- B. Defined as $\max(0, x)$
- C. Always suffers vanishing gradients
- D. Linear activation

Correct Answer: B

Detailed Solution: $\text{ReLU}(x) = \max(0, x)$.

Question 6

Which of the following statements about the learning rate for gradient descent are true?

1. A very small learning rate may result in slow convergence.
2. A very large learning rate may cause training to diverge.
3. Increasing the learning rate always improves performance.

- A. 1 only
- B. 2 only
- C. 1 and 2 only
- D. 1, 2 and 3

Correct Answer: C

Detailed Solution: Small learning rates make training slow, while excessively large learning rates may overshoot the minimum. Increasing the learning rate does not always improve performance.

Question 7

Which component introduces non-linearity into a neural network?

- A. Weights
- B. Bias
- C. Activation functions
- D. Inputs

Correct Answer: C

Detailed Solution: Activation functions introduce non-linearity.

Question 8

Which neural network architecture is most suitable for sequential data?

- A. Convolutional Neural Networks
- B. Perceptron
- C. Recurrent Neural Networks
- D. Linear Regression

Correct Answer: C

Detailed Solution: RNNs are designed for sequential data.

Question 9

A fully connected neural network has:

- An input layer with 4 neurons
- One hidden layer with 5 neurons
- An output layer with 2 neurons

Assume every hidden neuron is connected to every input neuron and every output neuron is connected to every hidden neuron.

Ignoring bias terms, how many trainable weights are there in the network?

- A. 20
- B. 18
- C. 30
- D. 40

Correct Answer: C

Detailed Solution:

Weights between input and hidden layer: $4 \times 5 = 20$

Weights between hidden and output layer: $5 \times 2 = 10$

Total number of weights: $20 + 10 = 30$

Question 10

Training loss decreases but validation loss increases. This indicates:

- A. Underfitting
- B. Overfitting
- C. Gradient Explosion
- D. Data Normalization

Correct Answer: B

Detailed Solution: This is a classic sign of overfitting.
